Slide Set #4

CLRM estimation

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Notes

Least Squares Method (LS)

Given an additive regression model:

$$\boldsymbol{y} = f(\boldsymbol{X}; \boldsymbol{\beta}) + \boldsymbol{\varepsilon}$$

note that $\boldsymbol{\varepsilon}$ is not observed, but it is function of observables and the unknown parameter

$$\boldsymbol{\varepsilon} = \boldsymbol{y} - f(\boldsymbol{X}; \boldsymbol{\beta})$$

LS method:

- assume the signal $f(\boldsymbol{X}; \boldsymbol{\beta})$ is much stronger than the error $\boldsymbol{\varepsilon}$.
- look for a $\boldsymbol{\beta}$ such that the "size" of $\boldsymbol{\varepsilon}$ is as small as possible
- size of $\boldsymbol{\varepsilon}$ is measured by some norm $\|\boldsymbol{\varepsilon}\|$

Notes

Ordinary Least Squares estimator (OLS)

OLS = LS with $\|\cdot\|_2$. Therefore the OLS objective function is

$$S(\boldsymbol{\beta}) = \|\boldsymbol{\varepsilon}\|_2^2 = \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} = (\boldsymbol{y} - f(\boldsymbol{X};\boldsymbol{\beta}))'(\boldsymbol{y} - f(\boldsymbol{X};\boldsymbol{\beta})),$$

and the OLS estimator $\boldsymbol{b}$ is defined as the optimal solution

$$\boldsymbol{b} = \underset{\boldsymbol{\beta}\in\mathbb{R}^K}{\arg\min}\, S(\boldsymbol{\beta})$$

For the linear model

$$S(\boldsymbol{\beta}) = \|\boldsymbol{\varepsilon}\|_2^2 = \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (y_i - \boldsymbol{x}_i'\boldsymbol{\beta})^2$$

Remark: $S(\boldsymbol{\beta})$ is nicely convex, therefore a minimum exists

Notes

Proposition: OLS estimator

The "unique" OLS estimator is $\boldsymbol{b} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$

To see this, first we introduce two simple matrix derivative rules:

1. Let $\boldsymbol{a}, \boldsymbol{b} \in R^p$ then
$$\frac{\partial \boldsymbol{a}'\boldsymbol{b}}{\partial \boldsymbol{b}} = \frac{\partial \boldsymbol{b}'\boldsymbol{a}}{\partial \boldsymbol{b}} = \boldsymbol{a}$$
2. Let $\boldsymbol{b} \in \mathbb{R}^p$, and let $\boldsymbol{A} \in \mathbb{R}^{p\times p}$ be symmetric, then
$$\frac{\partial \boldsymbol{a}' A\boldsymbol{b}}{\partial \boldsymbol{b}} = 2\boldsymbol{A}\boldsymbol{b} = 2\boldsymbol{b}'\boldsymbol{A}$$

Notes

Proof. Rewrite the LS objective function

$$\begin{aligned} S(\boldsymbol{\beta}) =& (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) \\ =& \boldsymbol{y}'\boldsymbol{y} - \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{y} - \boldsymbol{y}'\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta} \end{aligned}$$

Note that the transpose of a scalar is the scalar itself, then

$$\boldsymbol{y}'\boldsymbol{X}\boldsymbol{\beta} = (\boldsymbol{y}'\boldsymbol{X}\boldsymbol{\beta})' = \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{y}$$

so that we write

$$S(\boldsymbol{\beta}) = \boldsymbol{y}'\boldsymbol{y} - 2\boldsymbol{\beta}'(\boldsymbol{X}'\boldsymbol{y}) + \boldsymbol{\beta}'(\boldsymbol{X}'\boldsymbol{X})\boldsymbol{\beta} \tag{4.1}$$

Since $S(\cdot)$ is convex, there exists a minimum $\boldsymbol{b}$ which will satisfy the first order conditions

$$\left.\frac{\partial S(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}\right|_{\boldsymbol{\beta}=\boldsymbol{b}} = \boldsymbol{0}$$

Notes

By applying the previous derivative rules (1) and (2) to the 2nd and 3rd term of (4.1)

$$\frac{\partial S(\boldsymbol{b})}{\partial \boldsymbol{b}} = -2(\boldsymbol{X}'\boldsymbol{y}) + 2(\boldsymbol{X}'\boldsymbol{X})\boldsymbol{b} = \boldsymbol{0}$$

Which lead to the so called "normal equations"

$$(\boldsymbol{X}'\boldsymbol{X})\boldsymbol{b} = \boldsymbol{X}'\boldsymbol{y}$$

The matrix $\boldsymbol{X}'\boldsymbol{X}$ is square symmetric (see homeworks). Based on A3 with probability 1 $\boldsymbol{X}'\boldsymbol{X}$ is non singular, then $(\boldsymbol{X}'\boldsymbol{X})^{-1}$ exists, then the normal equation can be written as

$$(\boldsymbol{X}'\boldsymbol{X})^{-1}(\boldsymbol{X}'\boldsymbol{X})\boldsymbol{b} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} \quad \Longrightarrow \quad \boldsymbol{b} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$$

which proves the desired result ■

Notes

Formulation in terms of sample averages

It can be shown (see homeworks) that

$$\boldsymbol{X}'\boldsymbol{X} = \sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i' \qquad \text{and} \qquad \boldsymbol{X}'\boldsymbol{y} = \sum_{i=1}^{n} \boldsymbol{x}_i y_i$$

Define

$$\boldsymbol{S}_{\boldsymbol{xx}} = \frac{1}{n}\boldsymbol{X}'\boldsymbol{X} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i' \qquad \text{and} \qquad \boldsymbol{s}_{\boldsymbol{xy}} = \frac{1}{n}\boldsymbol{X}'\boldsymbol{y} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i y_i$$

Therefore $\boldsymbol{b} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$ can be written as

$$\begin{aligned}
\boldsymbol{b} &= \left(\frac{1}{n}\boldsymbol{X}'\boldsymbol{X}\right)^{-1}\frac{1}{n}\boldsymbol{X}'\boldsymbol{y} \\
&= \left(\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i y_i\right) \\
&= \boldsymbol{S}_{\boldsymbol{xx}}^{-1}\, \boldsymbol{s}_{\boldsymbol{xy}}
\end{aligned}$$

Notes

Once $\boldsymbol{\beta}$ is estimated via $\boldsymbol{b}$, the estimated error, also called "*residual*" is obtained as

$$\boldsymbol{e} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{b}$$

Fitted values, also called the predicted values, are

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{b}$$

so that $\boldsymbol{e} = \boldsymbol{y} - \hat{\boldsymbol{y}}$

Note that

$$\hat{y}_i = b_1 + b_2 x_{i2} + b_2 x_{i2} + \ldots \quad \text{for all} \quad i = 1, 2, \ldots, n$$

What is $\hat{y}_i$? $\hat{y}_i$ I's the estimated conditional expectation of Y for the when $X_1 = 1, X_2 = x_{i2}, \ldots, X_K = x_{iK}$

Notes

Algebraic/Geometric properties of the OLS

Proposition: orthogonality of residuals

The column space of $\boldsymbol{X}$ is orthogonal to the residual vector

Proof. Write the normal equations

$$\boldsymbol{X}'\boldsymbol{X}\boldsymbol{b} - \boldsymbol{X}'\boldsymbol{y} = \boldsymbol{0} \implies \boldsymbol{X}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b}) = \boldsymbol{0} \implies \boldsymbol{X}'\boldsymbol{e} = \boldsymbol{0}$$

Therefore for every column $\boldsymbol{X}_{\cdot k}$ (observed regressor) it holds true that the inner product $\boldsymbol{X}_{\cdot k}{}'\boldsymbol{e} = \boldsymbol{0}$. ■

Notes

Proposition: residuals sum to zero

If the linear model includes the constant term, then

$$\sum_{i=1}^{n} e_i = \sum_{i=1}^{n} (y_i - \boldsymbol{x}_i'\boldsymbol{b}) = 0$$

Proof. By assumption we have a liner model with constant/intercept term.That is

$$y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \ldots + \varepsilon_i$$

Therefore $\boldsymbol{X}_{\cdot 1} = \boldsymbol{1}_n = (1, , 1, \ldots, 1)'$. Apply the previous property the 1st column of $\boldsymbol{X}$

$$\boldsymbol{X}_{\cdot 1}{}'\boldsymbol{e} = \boldsymbol{1}'\boldsymbol{e} = \sum_{i=1}^{n} e_i = 0$$

and this proves the property ■

Notes

Proposition: Fitted vector is a projection

$\hat{\boldsymbol{y}}$ is the projection of $\boldsymbol{y}$ onto the space spanned by columns of $\boldsymbol{X}$ (regressors)

Proof.

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{b} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} = \boldsymbol{P}\boldsymbol{y}$$

It suffices to show that that $\boldsymbol{P} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$ is symmetric and idempotent.

$$\begin{aligned}
\boldsymbol{P}' &= \left(\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\right)' \\
&= \boldsymbol{X}\left((\boldsymbol{X}'\boldsymbol{X})^{-1}\right)'\boldsymbol{X}' \\
&= \boldsymbol{X}\left((\boldsymbol{X}'\boldsymbol{X})'\right)^{-1}\boldsymbol{X}' \\
&= \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' = \boldsymbol{P}
\end{aligned}$$

Therefore $\boldsymbol{P}$ is symmetric.

Notes

$$\begin{aligned}
\boldsymbol{P}\boldsymbol{P} &= \left(\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\right)\left(\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\right) \\
&= \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}(\boldsymbol{X}'\boldsymbol{X})(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' \\
&= \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' = \boldsymbol{P}
\end{aligned}$$

which shows that $\boldsymbol{P}$ is also idempotent, and this completes the proof ■

$\boldsymbol{P}$ it's called the influence matrix, because measures the impact of the observed ys on each predicted $\hat{y}_i$.

Elements of the diagonal of $\boldsymbol{P}$ are called leverages, because are the influence y_i on the the corresponding $\hat{y}_i$

Notes

Proposition: orthogonal decomposition

The OLS fitting decomposes the observed vector $\boldsymbol{y}$ in the sum of two orthogonal components

$$\boldsymbol{y} = \hat{\boldsymbol{y}} + \boldsymbol{e} = \boldsymbol{P}\boldsymbol{y} + \boldsymbol{M}\boldsymbol{y}$$

Remark: orthogonality implies that the individual contributions of each term of the decomposition of $\boldsymbol{y}$ are somewhat *well identified*.

Proof. First notice that

$$\boldsymbol{e} = \boldsymbol{y} - \hat{\boldsymbol{y}} = \boldsymbol{y} - \boldsymbol{P}\boldsymbol{y} = (\boldsymbol{I} - \boldsymbol{P})\boldsymbol{y} = \boldsymbol{M}\boldsymbol{y}$$

where $\boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{P})$. Therefore

$$\boldsymbol{y} = \hat{\boldsymbol{y}} + \boldsymbol{e} = \boldsymbol{P}\boldsymbol{y} + \boldsymbol{M}\boldsymbol{y}$$

It remains to show that $\hat{\boldsymbol{y}} = \boldsymbol{P}\boldsymbol{y}$ and $\boldsymbol{e} = \boldsymbol{M}\boldsymbol{y}$ are orthogonal vectors.

Notes

First note that $\boldsymbol{M}\boldsymbol{P} = \boldsymbol{P}\boldsymbol{M} = \boldsymbol{0}$, in fact

$$(\boldsymbol{I} - \boldsymbol{P})\boldsymbol{P} = \boldsymbol{I}\boldsymbol{P} - \boldsymbol{P}\boldsymbol{P} = \boldsymbol{0}$$

Moreover

$$\langle \boldsymbol{P}\boldsymbol{y}, \boldsymbol{M}\boldsymbol{y} \rangle = (\boldsymbol{P}\boldsymbol{y})'(\boldsymbol{M}\boldsymbol{y}) = \boldsymbol{y}'\boldsymbol{P}'\boldsymbol{M}\boldsymbol{y} = \boldsymbol{y}'\boldsymbol{P}\boldsymbol{M}\boldsymbol{y} = \boldsymbol{y}'\boldsymbol{0}\boldsymbol{y} = \boldsymbol{0}$$

and this completes the proof ■

$\boldsymbol{M} = \boldsymbol{I} - \boldsymbol{P}$ is called the residual maker matrix because it maps $\boldsymbol{y}$ into $\boldsymbol{e}$. It allows to write $\boldsymbol{e}$ in terms of the observables $\boldsymbol{y}$ and $\boldsymbol{X}$.

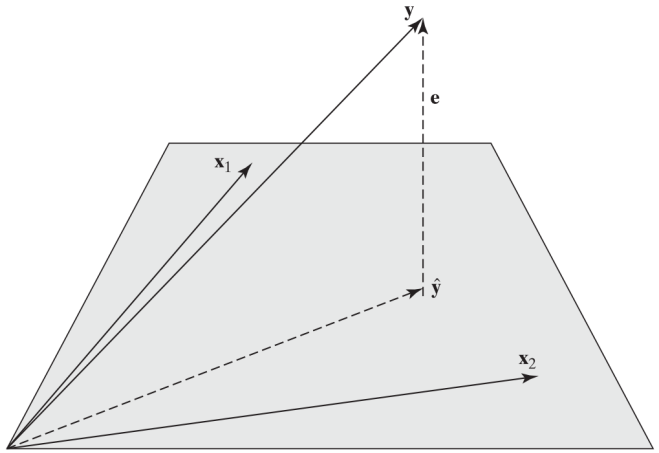
Properties:

- $\boldsymbol{M}$ is idempotent and symmetric (show it)
- $\boldsymbol{M}\boldsymbol{X} = \boldsymbol{0}$, in fact $\boldsymbol{M}\boldsymbol{X} = (\boldsymbol{I} - \boldsymbol{P})\boldsymbol{X} = \boldsymbol{X} - \boldsymbol{X} = \boldsymbol{0}$

Remark: it can be shown that this decomposition is also *unique* (a consequence of Hilbert projection theorem).

Notes

OLS Projection



Source: Greene, W. H. (2011) "*Econometric Analysis*" 7th Edition

Notes

Estimate of the variance of the error term

Min of the LS objective function

$$S(\boldsymbol{b}) = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b}) = \boldsymbol{e}'\boldsymbol{e}$$

This called "Residual sum of squares"

$$\text{RSS} = \sum_{i=1}^{n} e_i^2 = \boldsymbol{e}'\boldsymbol{e}$$

Note that

$$\boldsymbol{e} = \boldsymbol{M}\boldsymbol{y} = \boldsymbol{M}(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{M}\boldsymbol{\varepsilon}$$

and

$$\text{RSS} = \boldsymbol{e}'\boldsymbol{e} = (\boldsymbol{M}\boldsymbol{\varepsilon})'(\boldsymbol{M}\boldsymbol{\varepsilon}) = \boldsymbol{\varepsilon}'\boldsymbol{M}'\boldsymbol{M}\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}'\boldsymbol{M}\boldsymbol{\varepsilon}$$

Notes

Unbiased estimation of the error variance

$$s^2 = \frac{1}{n-K}\sum_{i=1}^{n} e_i^2 = \frac{\boldsymbol{e}'\boldsymbol{e}}{n-K} = \frac{\text{RSS}}{n-K}$$

SER = "*standard error of the regression*" $= s$

Notes

Estimation error decomposition

The sampling estimation error is given by $\boldsymbol{b} - \boldsymbol{\beta}$, now

$$\begin{aligned}
\boldsymbol{b} - \boldsymbol{\beta} &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} - \boldsymbol{\beta} \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) - \boldsymbol{\beta} \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}(\boldsymbol{X}'\boldsymbol{X})\boldsymbol{\beta} + (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon} - \boldsymbol{\beta} \\
&= \boldsymbol{\beta} + (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon} - \boldsymbol{\beta} \\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon}
\end{aligned}$$

The bias is the expected estimation error:

$$\text{Bias}(\boldsymbol{b}) = \mathrm{E}[\boldsymbol{b} - \boldsymbol{\beta}]$$

Notes

TSS = total sum of squares

Let $\bar{y}$ be the sample average of the observed $y_1, y_2, \ldots, y_n$:

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i,$$

and let $\bar{\boldsymbol{y}} = (\underbrace{\bar{y}, \bar{y}, \ldots, \bar{y}}_{n \text{ times}})'$. We can also write $\bar{\boldsymbol{y}} = \bar{y}\mathbf{1}_n$

TSS = the deviance (variability) observed in the independent variable y

$$\text{TSS} = \sum_{i=1}^{n}(y_i - \bar{y})^2 = (\boldsymbol{y} - \bar{\boldsymbol{y}})'(\boldsymbol{y} - \bar{\boldsymbol{y}})$$

This is a variability measure, because it computes the squared deviations of y from its observed unconditional mean.

Notes

ESS = explained sum of squares

ESS = the overall deviance of the predicted values of y wrt to the unconditional mean of y

$$\text{ESS} = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 = (\hat{\boldsymbol{y}} - \bar{\boldsymbol{y}})'(\hat{\boldsymbol{y}} - \bar{\boldsymbol{y}})$$

At first look this is not exactly a measure of variability (why?). But it turns out that another property of the OLS is that

$$\frac{1}{n}\sum_{i=1}^{n}\hat{y}_i = \frac{1}{n}\sum_{i=1}^{n} y_i$$

Notes

TSS decomposition and goodness of fit

It can be shown (we don't do this here) that

$$\text{TSS} = \text{ESS} + \text{RSS}$$

From the previous decomposition we get a famous (and misused) goodness of fit statistic

$$\begin{aligned} R^2 &= \frac{\text{ESS}}{\text{TSS}} \\ &= 1 - \frac{\text{RSS}}{\text{TSS}} \end{aligned}$$

R^2 is the portion of deviance observed in the y that is explained by the linear model. This is also called *coefficient of determination*.

Notes

Problems with R^2

- Increases by adding more regressors. For this reason its better to look at the so-called adjusted R^2 (for the degrees of freedom) which is computed as follows:

$$\overline{R}^2 = 1 - \frac{\text{RSS}/(n-K)}{\text{TSS}/(n-1)}$$

- $R^2 \in [0, 1]$ only if the constant term is included in the model. So when you estimate without intercept don't be scared if you get $R^2 < 0$
- An extremely large R^2 is pathological, guess why!

Notes